Offline Safe Reinforcement Learning for eMBB/URLLC Resource Allocation in O-RAN using Real Network Traces

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***Abstract*—In the Open Radio Access Network (O-RAN), xApps in the Near-Real-Time RAN Intelligent Controller jointly configure Physical Resource Block (PRB) slicing and Medium Access Control (MAC) scheduling between enhanced Mobile Broadband (eMBB) and Ultra-Reliable Low-Latency Communications (URLLC) slices, whose objectives conflict. Reliable learning-based control requires understanding how configuration affects service-level objectives, especially under realistic traces rather than synthetic simulators. We propose an audit-first methodology for safe O-RAN control combining trace-driven safety modeling, systematic identification of the dominant SLA-violation sources, and a practical deployment-oriented framework for offline policy learning and evaluation. On commercial O-RAN traces with application-layer measured URLLC latency, our audit reveals a clear separation of control effects: PRB allocation mainly sets eMBB throughput, whereas the MAC scheduler dominates URLLC tail latency (99th-percentile ≈ 18 ms under Round-Robin versus more than 130 ms under Proportional-Fair). We then build a load-conditioned empirical URLLC safety model and integrate it into a constrained offline RL pipeline based on conservative value estimation and Lagrangian optimization. Trained on roughly 2.25×10^5 transitions and evaluated on one held-out deployment cluster without online exploration, it reduces URLLC SLA violations by 25–47% over unconstrained baselines for only a 1–2% eMBB throughput loss, validated by three off-policy estimators (matching, tabular OPE, and Fitted Q Evaluation). These results show that trace-driven safety modeling and audit-first evaluation support practical, data-grounded O-RAN control without unsafe online experimentation.**

***Index Terms*—O-RAN, offline reinforcement learning, safe RL, Conservative Q-Learning, resource allocation, network slicing, URLLC, eMBB.**

I. Introduction

The rapid diversification of mobile applications is driving the evolution of wireless networks beyond 5G toward 6G, where ubiquitous AI services are expected to play a central role [1]. These networks must accommodate heterogeneous service requirements, including enhanced Mobile Broadband (eMBB), which targets high throughput, and Ultra-Reliable Low-Latency Communications (URLLC), which requires stringent latency and reliability. Because both contend for shared Physical Resource Blocks (PRBs), jointly configuring slicing and scheduling is a central control problem in O-RAN, where AI-based xApps in the Near-Real-Time RAN Intelligent Controller (Near-RT RIC) enable closed-loop RAN adaptation using telemetry collected over the E2 interface [2].

Reinforcement Learning (RL) is well suited to this task, yet RL-based O-RAN controllers face obstacles near deployment: learning through direct interaction can expose the network to exploratory configurations that violate URLLC Service Level Agreements (SLAs), while simulation-based training relies on synthetic service models of uncertain fidelity. *Offline RL* sidesteps both issues, learning policies from previously collected traces without live interaction and admitting explicit URLLC safety constraints during training. It does not, however, guarantee a useful or safe policy: its effectiveness hinges on whether the traces contain sufficient action-dependent variation and a measurable safety signal—properties that must be established in real O-RAN data before any policy optimization.

We therefore begin by auditing the WINES Open RAN Commercial Traffic Twinning dataset [3], which provides PHY-, MAC-, and application-layer KPMs from twinned real-world cellular traffic. In particular, we derive one-way URLLC packet latency from the application-layer measurements and use it as the safety signal. This enables us to ground the safety analysis in observed application-layer behavior rather than a synthetic service model. The audit yields an initially counterintuitive finding: once offered load is controlled and the URLLC slice is isolated, varying the PRB split strongly affects eMBB throughput but barely moves URLLC latency, whereas the MAC scheduler choice (round-robin vs. proportional-fair) shifts URLLC tail latency by up to a factor of eight. The safety/throughput dilemma thus lives primarily in the scheduler dimension—a structure we make explicit and then exploit. Our contributions are:

- **Trace-driven safety modeling.** We build a fully offline constrained-MDP formulation of the eMBB/URLLC trade-off directly from commercial O-RAN traces using application-layer latency measurements, with no synthetic service model, and a load-aware empirical cost model estimated from the traces themselves.
- **Audit-first methodology.** We develop a systematic data audit covering data quality, temporal dynamics, action separability, and configuration/load confounding. It identifies scheduler selection, not the PRB split, as the primary URLLC latency lever—the structural finding that drives the whole design.
- **Practical deployment framework.** We instantiate the

pipeline with an adaptive safe CQL-Lagrangian agent and evaluate it on one held-out deployment cluster using three independent off-policy estimators: action matching, tabular OPE, and Fitted Q Evaluation (FQE). The resulting estimates indicate a 25–47% reduction in URLLC SLA violations at a 1–2% eMBB throughput cost, without online exploration.

The remainder of the paper reviews related work in Section II, formulates the constrained problem in Section III, audits the dataset in Section IV, presents the method in Section V, reports the results in Section VI, and discusses limitations and conclusions in Section VII.

II. Related Work

As heterogeneous services—eMBB, URLLC, and mMTC—with conflicting requirements share one interface, static allocation becomes inadequate [4], and O-RAN addresses this through the Near-RT RIC, whose xApps take closed-loop radio decisions on a 10 ms–1 s timescale. RL is widely applied here—value-based and policy-gradient scheduling and slice management [5], multi-agent xApp coordination [**?**], and dynamic multi-slice frameworks such as DORA [6]—consistently improving throughput or spectral efficiency, but assuming online interaction with a simulator or live network and leaving the pre-deployment offline setting unaddressed.

Offline RL instead learns from a fixed historical dataset, appealing to operators who cannot afford exploratory actions on production networks [7]. Its central difficulty is *distribution shift*: for rarely observed actions the learned Q-values grow overly optimistic and the policy diverges. Conservative Q-Learning (CQL) counters this by penalizing out-of-distribution action values while preserving those supported by the data [8]. The few historical-data-driven networking applications, e.g. predictive slicing [9], optimize throughput or spectral efficiency without explicit service-level safety constraints.

Since a throughput-maximizing policy may repeatedly exceed latency budgets, URLLC SLAs call for a safety-aware formulation, naturally expressed as a Constrained Markov Decision Process (CMDP) [10] and solved via Lagrangian relaxation, e.g. PID-Lagrangian control [11]. In networking, SLA-aware O-RAN slicing schemes include SafeSlice and CVaR-constrained allocation [12], [13], but these define the cost over latency, loss, or energy and require online interaction to estimate violations—limiting their use where exploratory actions may breach URLLC SLAs.

Existing O-RAN RL approaches are thus largely online, simulator-based, and throughput-oriented, and ill-suited to learning safety constraints from rare violations. We close this gap with a fully offline CMDP derived from commercial traces, combining a load-aware empirical cost model with adaptive Lagrangian optimization; while the algorithm itself is not novel, our contribution lies in its trace-grounded formulation, load-aware empirical safety model, and rigorous off-policy evaluation.

III. System Model and Problem Formulation

A. O-RAN Slicing and Scheduling Scenario

We consider a base station serving N_{UE} User Equipments (UEs) over a channel with bandwidth B and N_{PRB} Physical Resource Blocks (PRBs), partitioned between two slices with conflicting QoS objectives: an eMBB slice maximizing downlink throughput and a URLLC slice subject to a stringent latency SLA. The base station exports PHY/MAC Key Performance Measurements (KPMs) over the E2 interface to a Near-RT RIC hosting the resource-allocation xApp. At each control interval, the KPMs are aggregated into the system state, the Safe-CQL xApp selects an action, and the O-DU MAC scheduler enforces it, giving the control loop $s_t \to a_t \to \text{O-DU} \to s_{t+1}$.

B. Offline Dataset

We use the WINES Open RAN Commercial Traffic Twinning dataset [3], collected on the Colosseum O-RAN digital twin [14]: real LTE traces are twinned and replayed through the MGEN traffic generator [15], logging PHY/MAC KPMs together with *application-layer* packet latency. Measurements span multiple deployment conditions, slicing configurations, and MAC schedulers; a subset is retained per the data-quality and confounding criteria of Sec. IV, and one deployment unit is held out entirely to test generalization to an unseen condition. Each URLLC packet carries send and receive timestamps, whose difference gives the per-packet one-way latency.

Preprocessing transforms the raw traces into a constrained offline dataset

$$\mathcal{D} = \left\{(s_i, a_i, r_i, c_i, s_i')\right\}_{i=1}^{N_{\mathcal{D}}}, \tag{1}$$

with $N_{\mathcal{D}}$ valid transitions after filtering. Transitions are never formed across reservation or experiment boundaries: s_i' belongs to the following window within the same trajectory, whose last window is marked terminal. Trajectories from the held-out condition never appear in training.

C. Constrained MDP Formulation

We formulate the problem as a constrained MDP $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, C, \gamma)$ with unknown kernel P, reward and cost R, C, and discount factor $\gamma \in [0, 1)$. Each decision step is a non-overlapping 5 s window. At the ≈ 4 Hz KPM rate each window contains approximately 20 samples per UE—enough to average out sub-second fading and yield a reliable violation estimate, yet short enough to track the traffic dynamics driving the trade-off.

1) State space: The state $s_t = [x_t, h_t] \in \mathbb{R}^{12}$ combines the current situation x_t and a short history h_t:

$$x_t = \left[\, n_t,\ \overline{\mathrm{rsrp}}_t,\ \overline{\mathrm{snr}}_t,\ \overline{\mathrm{mcs}}_t,\ \overline{\mathrm{bler}}_t,\ b_t,\ \rho_t,\ \ell_t \,\right], \tag{2}$$

$$h_t = \left[\, \rho_{t-1},\ \rho_{t-2},\ \ell_{t-1},\ \ell_{t-2} \,\right]. \tag{3}$$

These are trace-derived radio and traffic indicators: the active-UE count n_t; eMBB radio metrics $\overline{\mathrm{rsrp}}_t$, $\overline{\mathrm{snr}}_t$, $\overline{\mathrm{mcs}}_t$, $\overline{\mathrm{bler}}_t$ over $\mathcal{U}_{\mathrm{e}}$; buffer occupancy b_t; eMBB throughput ρ_t and URLLC median latency ℓ_t; and their one- and two-step histories, which let the policy track demand trends without a recurrent

TABLE I: Joint PRB-allocation and scheduler action space.

eMBB/URLLC PRBs	Action a	
	RR	PF
9/41	0	1
21/29	2	3
30/20	4	5
39/11	6	7

architecture. Each variable is z-scored on training-split statistics only. The state excludes the current PRB split and scheduler: as the decision variables, embedding them would confound situation with action and prevent the policy from departing from the logged configuration.

2) Action space: The action jointly sets the PRB split and MAC scheduler,

$$a_t = (p_t, m_t), \quad p_t \in \{9, 21, 30, 39\}, \quad m_t \in \{\mathrm{RR}, \mathrm{PF}\}, \tag{4}$$

giving $|\mathcal{A}| = 8$ discrete actions (Table I), where p_t is the eMBB-slice PRB count from the four retained slicing configurations. The fifth configuration (no PRBs reserved for URLLC) is excluded as a degenerate, load-confounded case.These actions are exactly those present in the traces, so the policy is optimized and evaluated only over decisions with real data support.

3) Reward: The reward is the aggregate window-averaged downlink throughput of the eMBB slice. Let $\mathcal{U}_\mathrm{e}$ denote the set of eMBB UEs of windows t, $N_{u,t}$ the number of valid samples for UE u, and $d_u(k)$ is the downlink bitrate of UE u at sample k. We define the reward as follows:

$$r_t = 10^{-6} \sum_{u \in \mathcal{U}_\mathrm{e}} \bar{d}_{u,t}, \qquad \bar{d}_{u,t} = \frac{1}{N_{u,t}} \sum_{k=1}^{N_{u,t}} d_u(k), \tag{5}$$

where r_t is reported in Mbps and d_u in bit/s. Averaging each UE bitrate before aggregation avoids giving disproportionate weight to UEs with more samples, while summing across UEs capture the aggregate throughput delivered by the eMBB slide. In the retained dataset, r_t ranges from 0 to 23.5 Mbps, with a mean of 7.65 Mbps. Note that the reward is computed from measured KPMs rather than predicted by a performance model.

4) Cost: The safety cost is the per-window URLLC violation rate over URLLC-slice packets only—the fraction whose one-way latency ℓ_i exceeds $L_{\max}$

$$c_t = \frac{1}{N_t^\mathrm{u}} \sum_{i=1}^{N_t^\mathrm{u}} \mathbf{1}(\ell_i > L_{\max}) \in [0, 1]. \tag{6}$$

Restricting the cost to the URLLC slice is essential: the state's median latency ℓ_t captures central behavior, while this tail-sensitive rate defines safety. The mean per-window cost is 0.053 and 33% of windows contain a violation—dense enough to constrain directly. We also report the URLLC 99th-percentile latency as a tail-aware secondary metric.

5) Objective: The agent maximizes discounted eMBB return under a bound on the URLLC violation rate:

$$\max_\pi \underbrace{\mathbb{E}_\pi\Big[\textstyle\sum_{t=0}^{\infty} \gamma^t r_t\Big]}_{J_R(\pi)} \quad \text{s.t.} \quad \underbrace{\mathbb{E}_\pi\big[c_t\big]}_{J_C(\pi)} \le d, \qquad d = 0.03. \tag{7}$$

The asymmetry is deliberate: J_R is discounted because throughput accrues over decisions, whereas J_C is an expected per-window rate, since a decision already fixes the violation rate within its 5 s window and the operator only requires the average to stay below $d = 0.03$. We set $\gamma = 0.7$: the configuration is fixed per reservation with near-immediate effect, so a long horizon drives the Bellman target toward $r/(1-\gamma)$ while bootstrapping on itself—empirically the loss diverges at $\gamma = 0.99$ but converges at $\gamma = 0.7$. We solve (7) by Lagrangian relaxation with multiplier $\lambda \ge 0$,

$$\mathcal{L}(\pi, \lambda) = J_R(\pi) - \lambda\big(J_C(\pi) - d\big), \tag{8}$$

deferring the cost estimate and adaptive optimization of λ to Sec. V-D.

IV. Dataset Audit

We audit the traces along four axes—quality, temporal dynamics, action separability, and configuration/load confounding—to verify (rather than assume) an exploitable decision structure and measurable safety signal, establishing offline RL's feasibility and yielding the paper's central scheduler finding.

A. Data Quality and Experimental Plan

The dataset is a controlled plan—three deployment clusters, five slicing configurations, and two schedulers (30 configurations across $1{,}116$ reservations of up to 8 UEs). Parsing per-UE PHY/MAC KPMs and application-layer MGEN logs into 5 s windows yields $225{,}313$ transitions ($\approx 0.01\%$ missing after filtering unattached-UE records). The volume-balanced clusters ($335/360/421$ reservations) enable a clean leave-one-cluster-out protocol: holding out cluster 1 and training on clusters 2–3, all results measure generalization to one unseen cluster.

B. Action Separability: Two Distinct Levers

The action combines two sub-decisions—PRB split and scheduler—that act on different objectives.

a) PRB split controls throughput, not latency: Controlling for load (8-UE regime) and isolating URLLC-slice packets, eMBB throughput rises monotonically with eMBB PRBs (Pearson $+0.73$), while its correlation with URLLC latency is negligible ($+0.02$): even down to 11 PRBs, the light URLLC traffic is comfortably served. The PRB split is thus a throughput lever, not a safety lever.

b) The scheduler controls latency: The tension resides in the scheduler. On the URLLC slice, RR yields a 99th-percentile latency of ≈ 18 ms whereas PF ranges from 86 to 137 ms—up to eightfold, roughly sixfold on average—consistently across all four configurations (Mann–Whitney U, $p < 10^{-4}$). RR is not dominated on throughput either, matching or exceeding PF at every split, so PF sacrifices the temporal regularity

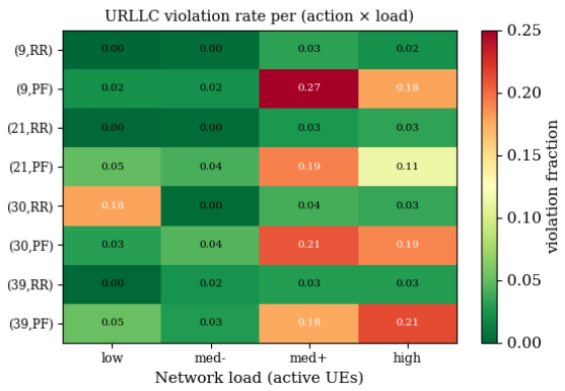


Fig. 1: URLLC SLA violation rate as a function of network load and selected actions.

URLLC needs with no compensating gain. This is the structural finding underpinning our formulation: the scheduler governs the URLLC SLA, and PF is the tempting-but-unsafe action a safe agent must learn to constrain.

C. Confounding and Load-Dependent Cost

Configurations are not sampled uniformly across load: the degenerate $50/0$ configuration (no URLLC PRBs) appears almost only at low load ($\leq$ 4 UEs), configurations 1–4 almost only at the 8-UE regime, with no overlap. Treating the configuration as freely selectable would thus conflate allocation with load, so we exclude $50/0$ and restrict the action space to configurations 1–4, where all PRB splits occur under comparable load.

The cost also depends on load. Fig. 1 reports the empirical violation rate per (action, load) pair: RR stays safe across all loads (typically $< 6\%$), whereas PF degrades sharply, reaching 19–27% at medium-to-high load—so avoiding PF matters most when congested, motivating an empirical per-(action, load) cost estimate.

V. Offline Safe Reinforcement Learning

We propose a trace-driven pipeline that turns an audited safety signal into a deployable constrained policy. Within it we benchmark three offline agents of increasing capability—Behavior Cloning (BC), Conservative Q-Learning (CQL), and an adaptive safe CQL-Lagrangian agent—all learning exclusively from logged transitions, so no unsafe configuration is ever explored during training.

We formulate the task as sequential RL rather than supervised learning or a contextual bandit for two reasons. First, the logged actions are the behavior policy's choices—several of them unsafe—not ground-truth labels, so imitating them reproduces the violations we seek to remove. Second, although the cost is near-immediate, the demand conditioning it evolves within a reservation, so the state carries temporal history and the value function bootstraps over successive windows—structure a one-shot bandit ignores. The short horizon $\gamma = 0.7$ is the deliberate consequence of this near-immediate effect, not a contradiction of it (below).

A. Behavior Cloning Baseline

BC trains a classifier $\pi_{\mathrm{BC}}(a \mid s)$ to predict the logged action by cross-entropy, with class weighting to offset the over-representation of RR. It reproduces the behavior policy and therefore inherits its high-cost PF decisions, serving both as a sanity check that the state carries predictive signal—held-out action-matching accuracy plateaus near 40%, with scheduler-choice accuracy (66%) exceeding PRB-level accuracy (58%)—and as the lower bound a useful offline method must beat.

B. Conservative Q-Learning

CQL learns a reward value function while avoiding the overestimation of out-of-distribution actions that plagues naive offline Q-learning [7], [8]. Over the eight discrete actions, built on a Deep Q-Network, it augments the temporal-difference objective with a conservative penalty that pushes down all Q-values and pulls up that of the logged action:

$$\mathcal{L}_{\mathrm{CQL}} = \underbrace{\mathbb{E}\Big[\big(Q(s,a)-y\big)^2\Big]}_{\text{Bellman error}} + \alpha\ \underbrace{\mathbb{E}\Big[\log \textstyle\sum_{a'} e^{Q(s,a')} - Q(s,a)\Big]}_{\text{conservative penalty}}, \tag{9}$$

with TD target $y = r + \gamma\,(1 - \mathrm{done})\max_{a'} Q_{\bar{\theta}}(s', a')$ and penalty weight $\alpha = 0.5$. To curb value divergence we use clipped double-Q learning (target = minimum of two critics), a Polyak-averaged target network ($\tau = 0.01$), and gradient-norm clipping.

$\gamma = 0.7$ (Fig. 2). Unconstrained CQL ($\lambda = 0$) shifts some mass from PF toward RR relative to BC, yet still selects PF often enough to leave its violation rate near that of imitation—so constraint-free value maximization is not safe.

C. Load-Aware Empirical Safety Model

Enforcing the SLA requires a cost model. Learning a cost critic $Q_c(s,a)$ by Bellman backup proved unstable exactly where it matters: because violations are rare and concentrated in a few (action, load) regions, the critic mis-estimated sparsely observed combinations—in one run labeling a high-violation PF action as safe—corrupting the constrained policy. We therefore replace it with an *empirical, tabular* model estimated on the training split. Exploiting the small action space and the strong action×load dependence of cost (Sec. IV-C), we tabulate the expected violation rate per (action, load-bin) pair,

$$\hat{c}(a,\beta) = \frac{1}{N_{a,\beta}} \sum_i c_i\, \mathbf{1}(a_i = a,\ \beta_i = \beta)\,, \tag{10}$$

where β bins the active-UE count n_t into three levels (low/medium/high) and cells with support $N_{a,\beta} < 20$ fall back to the action marginal $\bar{c}(a)$. This estimate is exact in expectation, far more robust than a function approximator under rare violations, and fully transparent—each configuration's risk is read directly from the measured traces (Fig. 1).

D. Adaptive Safe CQL-Lagrangian

The safe agent couples the reward critic Q_r with the empirical cost model through the Lagrangian relaxation of

Sec. III-C. Given $\lambda \geq 0$, it selects the action maximizing a penalized value,

$$\pi_\lambda(s) = \arg\max_a \ \big[\, Q_r(s,a) - \lambda\, \hat{c}(a,\beta) \,\big], \qquad (11)$$

so a larger λ steers the policy from high-cost PF toward safe RR. The cost is treated as immediate ($\gamma_c = 0$): a configuration's latency effect is confined to the current window and does not alter future exogenous traffic, consistent with the short reward horizon.

a) Adaptive multiplier: Crucially, λ is learned, not swept: it is optimized jointly with the Q-function by a primal–dual update. After each epoch it is adjusted by projected gradient ascent on the empirical constraint violation,

$$\lambda \leftarrow \big[\lambda + \eta_\lambda\big(\hat{J}_C(\pi) - d\big)\big]_+, \quad \hat{J}_C(\pi) = \frac{1}{N}\sum_i \hat{c}\big(\pi(s_i), \beta_i\big), \qquad (12)$$

rising while the policy's tabular cost exceeds the budget $d = 0.03$ and relaxing otherwise. Initialized small, λ grows monotonically and converges near 12.9 ± 0.6 as the constrained cost settles at the 3% budget (Fig. 3)—the behavior of a shadow price rather than a tuned constant. Training the constraint into the agent this way, rather than applying a penalty only at decision time, is what distinguishes Safe-CQL from a post-hoc penalty sweep.

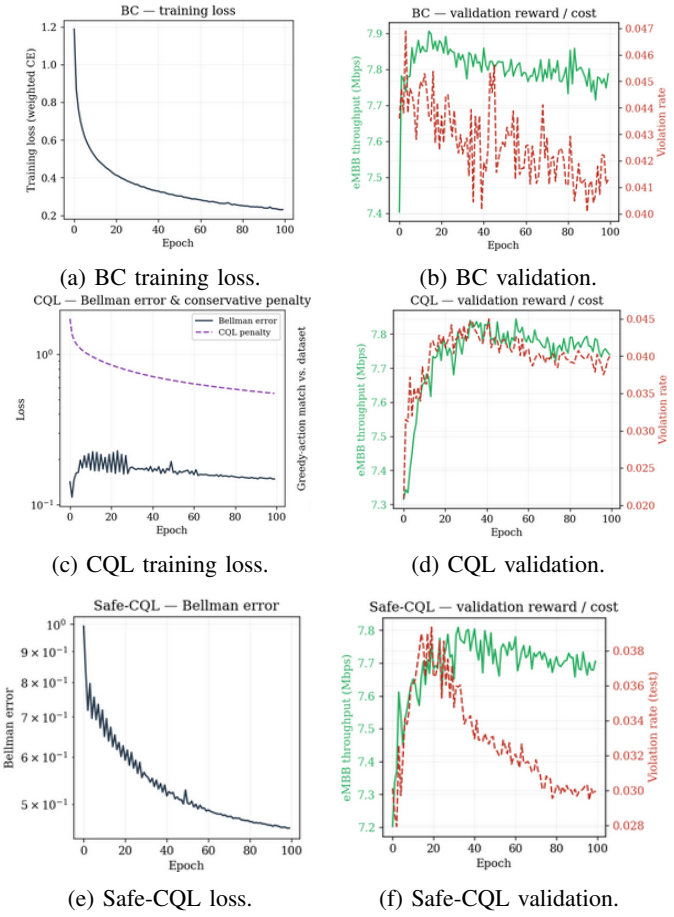
E. Off-Policy Evaluation

An offline policy cannot be run on the live network, so evaluation is off-policy; we use two primary estimators and one independent check and rely on their agreement. The *matching* estimator averages the true measured reward and cost on held-out transitions where the policy's action equals the logged one (unbiased but limited to matched coverage $\approx 40\%$). *Tabular OPE* reads the empirical (action, load-bin) reward and cost of the chosen action in every test state, covering the full distribution and producing the headline numbers of Table II. *Fitted Q Evaluation* (FQE), structurally independent, fits reward- and cost-value networks by Bellman backup; sharing no machinery with the table-based estimators, it guards against a shared bias, and because its values are discounted returns we use it to confirm the agent *ranking*, not the per-window budget. Since the configuration is fixed within each logged episode, none of the estimators observes a mid-episode action switch, so all are proxies for—not substitutes for—online evaluation.

VI. Experimental Results

A. Experimental Setup

All agents are trained on clusters 2 and 3 and evaluated on the held-out cluster 1, measuring generalization to one unseen deployment cluster. Features are normalized using training-split statistics only. BC and CQL are each trained with five random seeds and Safe-CQL with two; we report mean and standard deviation across seeds. Reward is eMBB throughput (Mbps) and cost the URLLC violation rate against $L_{\max} = 20$ ms. Training convergence is shown in Fig. 2, and the multiplier

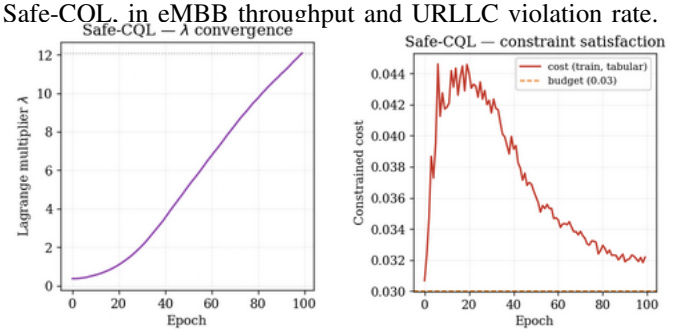


(a) BC training loss. (b) BC validation.

(c) CQL training loss. (d) CQL validation.

(e) Safe-CQL loss. (f) Safe-CQL validation.

Fig. 2: Training convergence and validation for BC, CQL, and Safe-CQL, in eMBB throughput and URLLC violation rate.



(a) Lagrange multiplier λ. (b) Constraint satisfaction.

Fig. 3: Safe-CQL primal–dual dynamics: λ converges near 12.9 (a) as the tabular training cost settles onto the $d = 0.03$ budget (b).

and constraint dynamics of the safe agent in Fig. 3: λ climbs monotonically while the tabular cost is driven onto the 3% budget.

B. Learned Policy Behavior

The gain comes not from allocating fewer PRBs to eMBB but from changing the scheduler preference—exactly the lever the audit identified. BC retains substantial mass on the high-cost PF configurations (shaded), reproducing the unsafe logged decisions; unconstrained CQL shifts somewhat toward RR but still selects PF in a non-negligible fraction of states; the safe

TABLE II: Offline safe RL benchmark on the held-out cluster.

	Matching OPE		Tabular OPE	
Method	Thr. (Mbps)	Cost	Thr. (Mbps)	Cost
BC (imitation)	7.76±0.10	0.041±0.002	**7.49±0.05**	0.057±0.002
CQL (λ=0)	**7.81±0.04**	0.040±0.002	7.40±0.03	0.049±0.002
CQL-safe	7.65±0.05	**0.030±0.000**	7.31±0.05	**0.030±0.001**

agent suppresses PF while preserving RR allocations across PRB levels—concentrating, e.g., on $(21, \mathrm{RR})$ and higher-PRB RR configurations. The constraint thus manifests as a learned, load-aware avoidance of proportional-fair scheduling, directly consistent with the cost structure of Fig. 1.

C. Main Results

Table II reports throughput and cost under the two primary estimators, and both agree on the safety ranking. BC and unconstrained CQL incur the highest violation rates (≈ 4.0–4.1% under matching, 4.9–5.7% under tabular OPE), whereas the safe agent reaches 3.0% under both—a $\approx 25\%$ reduction under matching and $\approx$ 39–47% under tabular OPE. FQE, evaluated independently, confirms the direction: its discounted cost value falls from 0.088 (CQL) to 0.076 (Safe-CQL). This lower cost is a direct consequence of the learned policy of Sec. VI-B: because the safe agent replaces PF with RR at medium-to-high load—the (action, load) cells carrying 19–27% violation rates in Fig. 1—while the PRB split (which the audit showed does not move URLLC latency) is left free to maximize throughput. The throughput price is therefore small: relative to unconstrained CQL the safe agent retains $\approx 98\%$ of throughput under matching and $\approx 99\%$ under tabular OPE. Across seeds the safe agent is stable, with standard deviations below 0.05 Mbps on throughput and 0.001 on cost;In short,*Safe-CQL reduces URLLC violations by* 25–47% *while sacrificing only* 1–2% *of eMBB throughput.*

D. Discussion

Three points stand out. First, unconstrained CQL is not by itself safe: it improves over imitation but retains enough PF decisions to violate the SLA at nearly the BC rate, and only the explicit constraint drives violations to the budget. Second, safety is nearly free because PF is not throughput-superior to RR on these traces—the safe agent avoids a *dominated* action, so 1–2% of throughput buys a 25–47% violation reduction. Third, this is achieved with no online interaction, learning to avoid the tempting-but-unsafe PF scheduler purely from historical traces—the decisive property for O-RAN, as it removes unsafe exploration on a production network while respecting the URLLC SLA.

VII. Limitations and Conclusion

Applying reinforcementl earning to a network dataset is often treated as an optimization exercise, but on real O-RAN traces the decisive question is whether the data supports the objective at all. Our audit answered this and reshaped the problem: because the safety dilemma lives in the scheduler rather than the PRB split, the safe policy's task was not to ration bandwidth but to avoid a tempting-but-unsafe scheduler—an action a throughput-maximizing agent has no reason to shun. This is why safety cost so little: the avoided action was never throughput-superior in the first place, so a 1–2% throughput concession bought a 25–47% violation reduction. The audit-first discipline behind this outcome is reusable independently of our specific agent: verifying data quality, temporal dynamics, action separability, and a measurable safety signal *before* committing to an algorithm guards against optimizing an objective the traces cannot support.

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